Canonical-Dual Sobolev Regularity:

A sharp shift obstruction and a positive low-regularity theorem

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17 August 2026

Status. Candidate substantial partial result, likely valid, pending human review. The concrete boundary-shearlet estimate at the intended integer Sobolev regularities remains open.

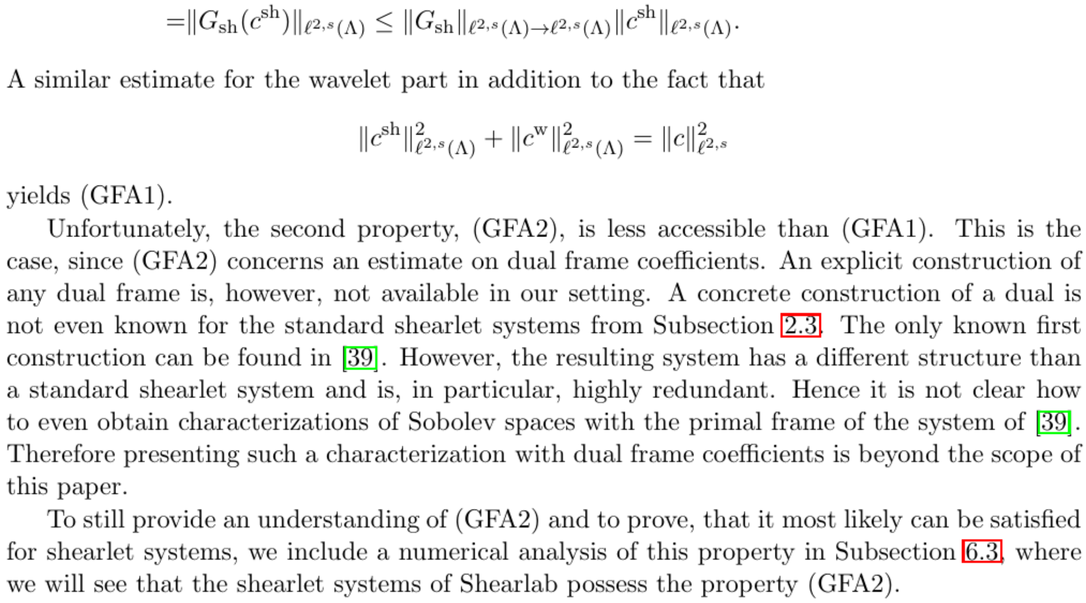
1 Source problem

Grohs, Kutyniok, Ma, Petersen, and Raslan construct boundary shearlet frames $\Phi = (\varphi_n)$ for $L^2(\Omega)$. Writing j_n for the scale, C and D for primal analysis and synthesis, and $S = DC$ for the L^2 frame operator, their desired Gelfand-frame estimates at regularity $s > 0$ are

$$\|Dc\|_{H^s(\Omega)} \lesssim \left\|(2^{j_n s} c_n)_n\right\|_2, \tag{GFA1}$$

$$\left\|(2^{j_n s}\langle f, S^{-1}\varphi_n\rangle)_n\right\|_2 \lesssim \|f\|_{H^s(\Omega)}. \tag{GFA2}$$

They prove a weighted primal-analysis characterization of $H^s(\Omega)$ and derive (GFA1) under localization assumptions. They explicitly leave (GFA2) without an analytic proof and provide numerical evidence instead [1, pp. 5, 18, and §6.3].



$$= \|G_{\mathrm{sh}}(c^{\mathrm{sh}})\|_{\ell^{2,s}(\Lambda)} \leq \|G_{\mathrm{sh}}\|_{\ell^{2,s}(\Lambda)\to\ell^{2,s}(\Lambda)} \|c^{\mathrm{sh}}\|_{\ell^{2,s}(\Lambda)}.$$

A similar estimate for the wavelet part in addition to the fact that

$$\|c^{\mathrm{sh}}\|^2_{\ell^{2,s}(\Lambda)} + \|c^{\mathrm{w}}\|^2_{\ell^{2,s}(\Lambda)} = \|c\|^2_{\ell^{2,s}}$$

yields (GFA1).

Unfortunately, the second property, (GFA2), is less accessible than (GFA1). This is the case, since (GFA2) concerns an estimate on dual frame coefficients. An explicit construction of any dual frame is, however, not available in our setting. A concrete construction of a dual is not even known for the standard shearlet systems from Subsection 2.3. The only known first construction can be found in [39]. However, the resulting system has a different structure than a standard shearlet system and is, in particular, highly redundant. Hence it is not clear how to even obtain characterizations of Sobolev spaces with the primal frame of the system of [39]. Therefore presenting such a characterization with dual frame coefficients is beyond the scope of this paper.

To still provide an understanding of (GFA2) and to prove, that it most likely can be satisfied for shearlet systems, we include a numerical analysis of this property in Subsection 6.3, where we will see that the shearlet systems of Shearlab possess the property (GFA2).

Figure 1: The source explains that (GFA2) is inaccessible because it concerns canonical-dual coefficients, and records only a numerical analysis [1, p. 18].

The first theorem below proves that the two available mapping properties do not imply (GFA2), even under very strong qualitative localization. The second theorem gives a positive analytic interval around $s = 0$.

2 Proof intuition

The obstruction is directional. A unilateral shift to a higher scale costs a factor 2^s in the weighted norm, while the backward shift gains the same factor. Choose a Riesz-basis synthesis operator $D = I - aR$. Its adjoint $D^* = I - aR^*$ controls the primal coefficients extremely well because it moves toward lower scales. Canonical-dual analysis, however, is D^{-1} and sums an infinite forward-shift tail. That tail belongs to the weighted space exactly below the threshold $a2^s = 1$.

For a general frame the frame operator S is nevertheless self-adjoint and invertible on the central L^2 space. Endpoint analysis and synthesis make S bounded at positive regularity, and self-adjointness supplies the negative endpoint by duality. Sneiberg's interpolation theorem then preserves invertibility on some open interval around zero. This proves (GFA2) there, although it gives no reason for invertibility to persist all the way to a prescribed integer exponent.

3 An exact phase transition

Let $\mathcal{H}_0 = \ell^2(\mathbb{N}_0)$ with standard basis (e_n), let $Re_n = e_{n+1}$, and for $r \geq 0$ put

$$\mathcal{H}_r = \left\{ x : \|x\|_r^2 = \sum_{n\geq 0} 2^{2rn}|x_n|^2 < \infty \right\}.$$

Use the same weighted space for coefficients and denote it by ℓ_r^2.

Theorem 1 (Bidiagonal Riesz-basis classification)**.** *Fix $0 < a < 1$, set $D = I - aR$, and define $\varphi_n = De_n$. Then (φ_n) is a Riesz basis of $\mathcal{H}_0$. For every $r \geq 0$:*

(i) *$D : \ell_r^2 \to \mathcal{H}_r$ is bounded, so* (GFA1) *holds;*

(ii) *its primal coefficients give an exact scale characterization,*

$$\|(\langle f, \varphi_n\rangle)_n\|_{\ell_r^2} = \|D^* f\|_r \asymp \|f\|_r;$$

(iii) *canonical-dual analysis is bounded $\mathcal{H}_r \to \ell_r^2$ if and only if*

$$a2^r < 1.$$

In particular, (GFA1) and the primal coefficient characterization do not imply (GFA2), even for a Riesz basis with tridiagonal Gramian and frame operator.

Proof. Because $\|aR\|_{\mathcal{H}_0} = a < 1$, the Neumann series makes D invertible on $\mathcal{H}_0$. Thus (De_n) is a Riesz basis, with ordinary frame bounds between $(1-a)^2$ and $(1+a)^2$.

Put $q = 2^r$. Directly from the weighted norm,

$$\|R\|_{\mathcal{H}_r} = q, \qquad \|R^*\|_{\mathcal{H}_r} = q^{-1}.$$

Consequently $\|Dc\|_r \leq (1 + aq)\|c\|_r$, proving (i). Primal analysis is $D^* = I - aR^*$, and

$$(D^*)^{-1} = \sum_{k\geq 0} a^k (R^*)^k$$

converges in $\mathcal{B}(\mathcal{H}_r)$ because $a/q < 1$. Equivalently,

$$(1 - a/q)\|f\|_r \leq \|D^* f\|_r \leq (1 + a/q)\|f\|_r,$$

which proves (ii).

The canonical dual basis is $((D^{-1})^* e_n)_n$, so its analysis map is D^{-1}. If $aq < 1$, the Neumann series $D^{-1} = \sum_{k\geq 0} a^k R^k$ converges on $\mathcal{H}_r$. Conversely,

$$D^{-1}e_0 = \sum_{k\geq 0} a^k e_k, \qquad \|D^{-1}e_0\|_r^2 = \sum_{k\geq 0} (aq)^{2k},$$

which diverges when $aq \geq 1$. This proves (iii). Since D is bidiagonal, D^*D and DD^* are tridiagonal. $\square$

Corollary 2 (Near-Parseval failure at fixed smoothness)**.** *Fix $r > 0$ and $\eta > 0$. There is a scale-labelled Riesz basis with ordinary L^2 frame bounds in $[(1-\eta)^2, (1+\eta)^2]$ for which (GFA1) and the primal characterization hold at r, but (GFA2) fails.*

Proof. Label e_n by scale $j_n = Ln$. The weight ratio is then $q = 2^{Lr}$. Choose L so large that $a = 2^{-Lr} < \min\{\eta, 1\}$. Theorem 1 applies with $aq = 1$. $\square$

4 A positive interval for every endpoint-stable frame

Let $(\mathcal{H}_t)_{-\sigma\leq t\leq\sigma}$ and $(\ell_t^2)_{-\sigma\leq t\leq\sigma}$ be exact complex interpolation Hilbert scales, with $\mathcal{H}_{-t} = \mathcal{H}_t'$ under the central $\mathcal{H}_0$ pairing. Standard Sobolev spaces and dyadically weighted ℓ^2 spaces have these properties.

Theorem 3 (Low-regularity Gelfand interval)**.** *Let Φ be a frame for $\mathcal{H}_0$, with analysis C, synthesis D, and frame operator $S = DC$. Suppose for some $\sigma > 0$ that*

$$C : \mathcal{H}_\sigma \to \ell_\sigma^2, \qquad D : \ell_\sigma^2 \to \mathcal{H}_\sigma$$

are bounded. Then there exists $\varepsilon \in (0, \sigma]$ such that, for every $0 \leq r < \varepsilon$, both (GFA1) and (GFA2) hold at exponent r. Equivalently, Φ is a Gelfand frame throughout a nonzero interval of regularities adjacent to L^2.

Proof. The hypotheses imply that $S = DC$ is bounded on $\mathcal{H}_\sigma$. Since S is self-adjoint for the $\mathcal{H}_0$ pairing, duality extends it boundedly to $\mathcal{H}_{-\sigma}$. It is boundedly invertible on $\mathcal{H}_0$ because Φ is an $\mathcal{H}_0$ frame.

Sneiberg's stability theorem for operators on complex interpolation scales now gives $\varepsilon > 0$ such that $S : \mathcal{H}_r \to \mathcal{H}_r$ is boundedly invertible whenever $|r| < \varepsilon$ [3]. Interpolation between the central and positive endpoints gives bounded maps $C : \mathcal{H}_r \to \ell_r^2$ and $D : \ell_r^2 \to \mathcal{H}_r$ for $0 \leq r \leq \sigma$. Hence GFA1 holds, and canonical-dual analysis satisfies

$$C_{\Phi^d} f = CS^{-1}f, \qquad \|C_{\Phi^d} f\|_{\ell_r^2} \leq \|C\| \, \|S^{-1}\| \, \|f\|_{\mathcal{H}_r}$$

for $0 \leq r < \varepsilon$. This is GFA2. $\square$

Remark 4. *Theorem 3 applies to a boundary-shearlet system whenever the source's endpoint coefficient characterization and localized synthesis bound are available at some $\sigma > 0$. It does not quantify ε from the published estimates and therefore need not reach $r = 1$.*

5 Consequences, limitation, and verification

Theorem 1 identifies the exact logical gap in the source's program. Boundedness of the Gramian on the weighted space, finite-band localization, and even near-tight L^2 frame bounds do not ensure that the canonical inverse respects the weight. A full boundary-shearlet proof needs a quantitative inverse-closed localization algebra whose decay dominates 2^{sj}; this is stronger than the componentwise weighted boundedness and the particular cross-decay estimate proved in the source. General inverse-closed localization results explain the appropriate route [4, 5], but verifying their hypotheses for the full hybrid index geometry was not achieved here.

All computations in Theorem 1 are exact geometric-series and operator-norm identities. The low-regularity theorem uses only frame operator self-adjointness, Hilbert-scale duality, interpolation of C, D, and Sneiberg stability. Eight focused routes were audited: the exact shift classification, near-Parseval strengthening, and low-regularity theorem succeeded; tight-frame reduction, direct implication, offset-Neumann, inverse-closed hybrid localization, and later-literature closure did not settle the intended integer-regularity problem.

The run's four lightweight indexes were searched by arXiv id, title, 'GFA2', 'boundary shearlet', 'canonical dual', and 'Gelfand frame'. A bounded arXiv/web search through 17 August 2026 found the source, its 2017 follow-up [2], and general localized-frame literature, but no analytic resolution for the concrete boundary-shearlet system or exact match for the shift phase transition. This is bounded evidence, not a priority claim.

Human-review recommendation. Promote for expert review as a substantial partial result. Check the shift orientation and canonical-dual identity in Theorem 1, and check the interpolation-couple hypotheses and use of Sneiberg's theorem in Theorem 3. Do not promote this packet as a full answer to the boundary-shearlet question without the missing inverse-closed localization estimate.

References

[1] P. Grohs, G. Kutyniok, J. Ma, P. Petersen, and M. Raslan, *Anisotropic Multiscale Systems on Bounded Domains*, arXiv:1510.04538 (2015; revised 2017), Adv. Comput. Math. 46 (2020), Art. 39.

[2] P. Petersen and M. Raslan, *Approximation properties of hybrid shearlet–wavelet frames for Sobolev spaces*, arXiv:1712.01047 (2017).

[3] I. Ya. Sneiberg, *Spectral properties of linear operators in interpolation families of Banach spaces*, Mat. Issled. 9 (1974), no. 2, 214–229.

[4] K. Gröchenig and A. Klotz, *Noncommutative approximation: inverse-closed subalgebras and off-diagonal decay of matrices*, Constr. Approx. 32 (2010), 429–466; arXiv:0904.0386.

[5] Q. Fang, C. E. Shin, and Q. Sun, *Polynomial control on weighted stability bounds and inversion norms of localized matrices on simple graphs*, arXiv:1909.08409 (2019).